LESSON 2.2

Represent and Describe Data in a Histogram

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can make and read a histogram to display data in intervals.

WHY THIS MATTERS The basketball league has 30 players and wants to display everyone's points-per-game average in a way that shows how the data is distributed. The raw averages are: 2, 5, 7, 8, 9, 10, 11, 11, 12, 13, 13, 14, 14, 15, 15, 16, 16, 17, 18, 18, 19, 20, 21, 22, 23, 24, 25, 28, 30, 32. Listing all 30 numbers is hard to read — a histogram will reveal the pattern!

REMEMBER

Represent and Describe Data in a Histogram

Histograms & Frequency Distributions A histogram is a bar graph that groups numbers into equal ranges called intervals. The height of each bar is the frequency: how many values fall in that interval. The bars touch because the intervals are continuous.

1. Set equal intervals along the horizontal axis with no gaps.
2. Count the frequency (how many data values fall into each interval).
3. Draw adjacent bars with heights equal to frequency

Words: **Histogram *(Histograma)*** **Frequency *(Frecuencia)*** **Interval *(Intervalo)*** **Distribution *(Distribución)*** **Variability *(Variabilidad)***

PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each word to what it means on a histogram.

_____ Frequency

_____ Interval

_____ Bars touch

(A) How many values are in an interval (the bar height)

(B) An equal-width range like 10–19

(C) Why a histogram has no gaps between bars

HINT Three vocab cards: one is a count, one is a range of values, and one explains the histogram's no-gap look.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Which of these is a statistical question?

(A) How many minutes are in one hour?

(B) How many days are in September?

(C) How many sides does a hexagon have?

(D) How many minutes did each student read last night?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

'How tall is our flagpole?' is NOT a statistical question. Why not?

(A) It is about height

(B) It uses numbers

(C) It has only one answer, so there is no variability

(D) Nobody can measure it

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A class collects this data on pets owned: 0, 2, 1, 0, 3. Which statistical question could these answers come from?

(A) How many students are in the class?

(B) What is the largest number of pets allowed?

(C) How many pets are in the store?

(D) How many pets does each student own?

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Counting Mistake

1. **Data (points per game)** 3, 7, 11, 13, 16, 22
2. **Intervals** 0–9, 10–19, 20–29
3. **Frequency count** 0–9: 2, 10–19: 2, 20–29: 1
4. **Conclusion** So the histogram bars have heights 2, 2, and 1.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Count the data values first: 3, 7, 11, 13, 16, 22 is 6 values, but the frequencies add to only 5. A value was missed.

CORRECT WORK

Correct answer: _____

6 Find the mistake. Explain it, then show the correct work.

Spot the Interval Mistake

1. **Data (assists per game)** 1, 4, 6, 9, 12, 15, 18
2. **Intervals the student chose** 0–5, 6–10, 12–18
3. **Frequency count** 0–5: 2, 6–10: 2, 12–18: 3
4. **Conclusion** The student drew three bars and said the histogram was complete.

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Check the width of each interval the student used. Are 0–5, 6–10, and 12–18 all the same size?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The tallest bar is the 10–19 interval with 16 players. What does the shape of this histogram tell you about scoring in the league?

SENTENCE STARTER Most players score between ___ and ___ points per game. The histogram is skewed ___ because ___. This tells me that ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Drag each player's points-per-game value into the correct interval bin (each bin is 10 wide).

Groups: **0–9 points** **10–19 points** **20–29 points**

_____ 6 points

_____ 8 points

_____ 13 points

_____ 17 points

_____ 21 points

_____ 25 points

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A soccer coach looks at a histogram of shots-per-game for the team: 0–2 shots: 1 player, 3–5 shots: 8 players, 6–8 shots: 10 players, 9–11 shots: 3 players, 12–14 shots: 1 player. The coach says, 'Most of my players take a good number of shots.' Which interval has the most players?

(A) 0–2 shots

(B) 3–5 shots

(C) 6–8 shots

(D) 9–11 shots

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A soccer coach looks at a histogram of shots-per-game for the team: 0–2 shots: 1 player, 3–5 shots: 8 players, 6–8 shots: 10 players, 9–11 shots: 3 players, 12–14 shots: 1 player. The coach says, 'Most of my players take a good number of shots.' How many players take between 3 and 8 shots per game?

(A) 8
(B) 10
(C) 18
(D) 23

PER 1

4 **Circle the letter of the best answer. Show how you know.**

A soccer coach looks at a histogram of shots-per-game for the team: 0–2 shots: 1 player, 3–5 shots: 8 players, 6–8 shots: 10 players, 9–11 shots: 3 players, 12–14 shots: 1 player. The coach says, 'Most of my players take a good number of shots.' Does the histogram support the coach's claim?

(A) No — most players take 0–2 shots
(B) No — the data is skewed right
(C) Yes — 18 of 23 players cluster in the middle intervals
(D) You cannot tell from a histogram

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Data: 4, 8, 9, 12, 14, 15, 17, 18, 19, 22, 24, 27, 31, 33. How many values fall in the 10–19 interval?

(A) 3
(B) 5
(C) 6
(D) 9

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

Frequencies: 0–9: 3, 10–19: 9, 20–29: 5, 30–39: 2. Which interval has the tallest bar?

(A) 0–9 with 3 players

(B) 10–19 with 9 players

(C) 20–29 with 5 players

(D) 30–39 with 2 players

SHOW YOUR WORK

EXPLAIN YOUR THINKING

The tallest bar is the 10–19 interval with 16 players. What does the shape of this histogram tell you about scoring in the league?

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